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VLSI - 第九周作业

1. $\begin{cases} D_0 = Q_3 \oplus Q_2 \\ D_1 = Q_0 \\ D_2 = Q_1 \\ D_3 = Q_2 \end{cases}$ $\begin{cases} Q_0^* = Q_3 \oplus Q_2 \\ Q_1^* = Q_0 \\ Q_2^* = Q_1 \\ Q_3^* = Q_2 \end{cases}$

| $Q_2^*$ | $Q_2^*$ | $Q_1^*$ | $Q_0^*$ | $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |

No.

Date

$0001 \to 0010 \to 0100 \to 1001 \to 0011 \to 0110 \to 1101 \to 1010 \to 0101 \to 1011 \to 0111 \to 1111 \to 1110 \to 1100 \to 1000 \to 0001$

$0000 \to 0000$

综上. 这是十五进制计数器. 不能自启动.

2, 74LS161. 异步清零 同步置数. 二进制.

两片之间

→即 5*16+2 = 82

当 $Q_3^{(2)} Q_2^{(2)} Q_1^{(2)} Q_0^{(2)} Q_3^{(1)} Q_2^{(1)} Q_1^{(1)} Q_0^{(1)} = \underbrace{0101}_{5}\ \underbrace{0010}_{2}$ 时 同步置数.

初始置数为0000 0000 即 "0"

计数从 0 - 82 且82有效

综上 此为83进制计数器 两片之间是16进制.

3.

| R | Y | G | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 |

$$\therefore\ R = Q_2'Q_1'Q_0 + Q_2Q_1'Q_0' + Q_2Q_1Q_0 \quad = \text{"}1+4+7\text{"}$$

$$Y = Q_2'Q_1Q_0' + Q_2Q_1'Q_0' + Q_2Q_1Q_0' \quad = \text{"}2+4+6\text{"}$$

$$G = Q_2'Q_1Q_0 + Q_2Q_1'Q_0' + Q_2Q_1'Q_0 \quad = \text{"}3+4+5\text{"}$$

~~138算中规模吗.~~ 算吧.